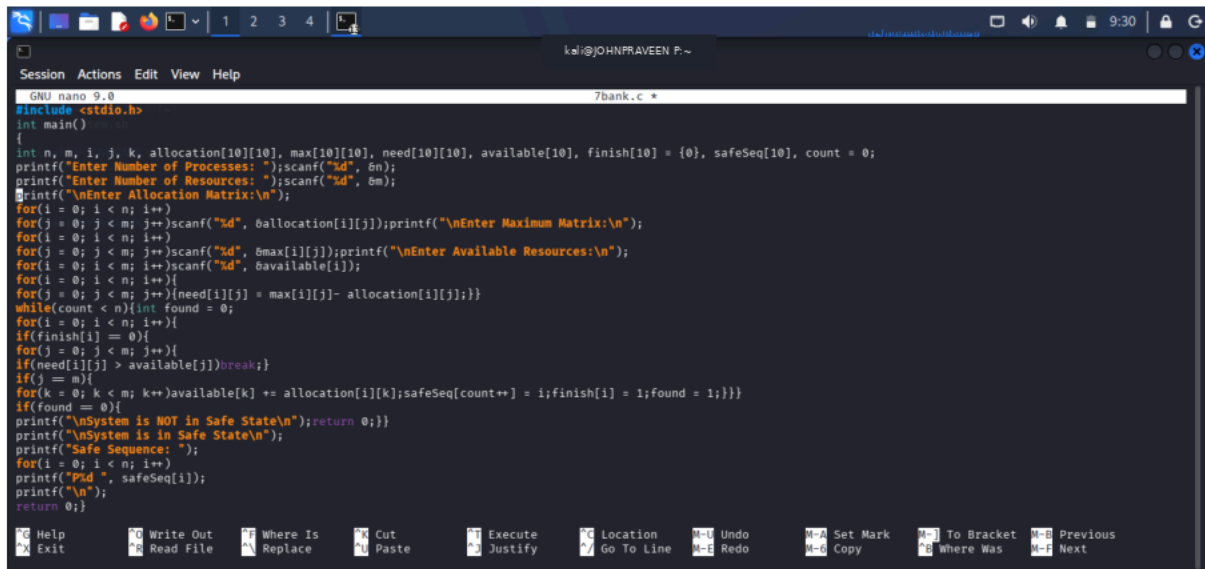
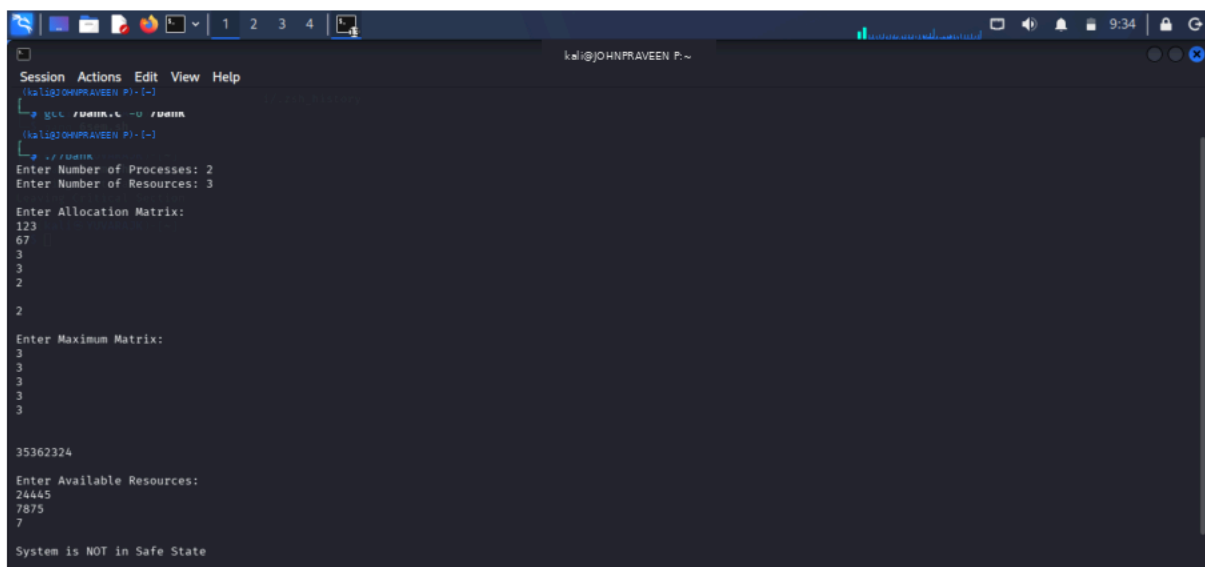


C PROGRAMMING :



```
GNU nano 9.0 7bank.c *
#include <stdio.h>
int main()
{
    int n, m, i, j, k, allocation[10][10], max[10][10], need[10][10], available[10], finish[10] = {0}, safeSeq[10], count = 0;
    printf("Enter Number of Processes: "); scanf("%d", &n);
    printf("Enter Number of Resources: "); scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++) scanf("%d", &allocation[i][j]); printf("\nEnter Maximum Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++) scanf("%d", &max[i][j]); printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++) scanf("%d", &available[i]);
    for(i = 0; i < n; i++){
        for(j = 0; j < m; j++){
            need[i][j] = max[i][j] - allocation[i][j];
        }
    }
    while(count < n){
        int found = 0;
        for(i = 0; i < n; i++){
            if(finish[i] == 0){
                for(j = 0; j < m; j++){
                    if(need[i][j] > available[j]) break;
                }
                if(j == m){
                    for(k = 0; k < m; k++) available[k] += allocation[i][k]; safeSeq[count++] = i; finish[i] = 1; found = 1;
                }
            }
            if(found == 0){
                printf("\nSystem is NOT in Safe State\n"); return 0;
            }
            printf("\nSystem is in Safe State\n");
            printf("Safe Sequence: ");
            for(i = 0; i < n; i++)
                printf("P%d ", safeSeq[i]);
            printf("\n");
            return 0;
        }
    }
```

OUTPUT :



```
(kali@JOHNPRAVEEN P:~)
└─$ gcc 7bank.c -o 7bank
(kali@JOHNPRAVEEN P:~)
└─$ ./7bank
Enter Number of Processes: 2
Enter Number of Resources: 3

Enter Allocation Matrix:
123
67
3
3
2
2

Enter Maximum Matrix:
3
3
3
3
3

35362324

Enter Available Resources:
24445
7875
7

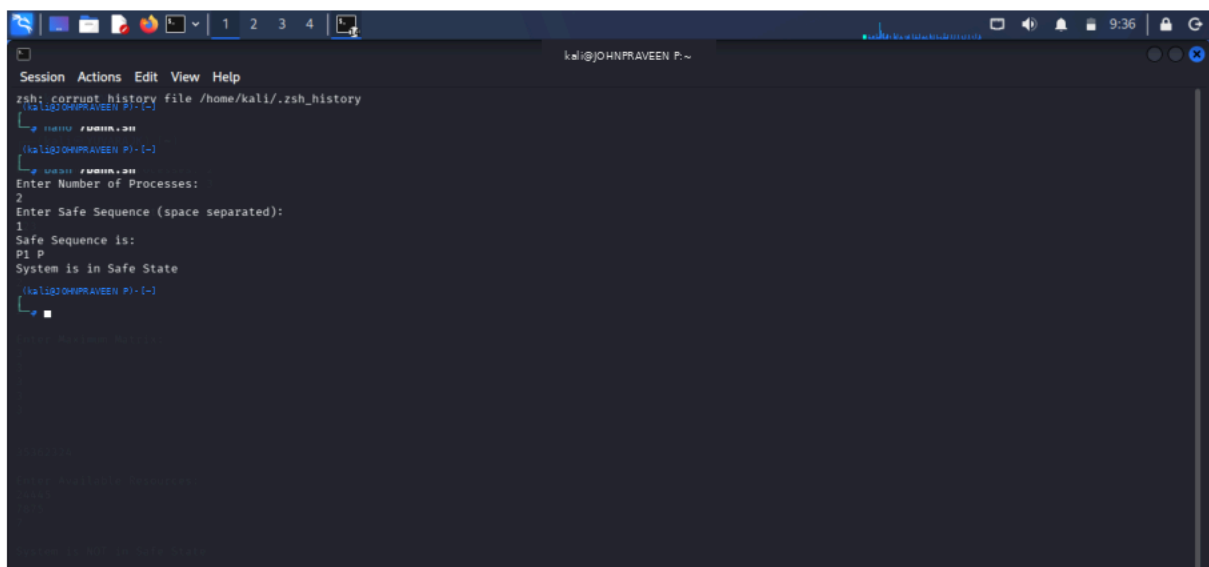
System is NOT in Safe State
```

SHELL PROGRAMMING :



```
GNU nano 9.0 7bank.sh
#!/bin/bash
echo "Enter Number of Processes:"
read n
echo "Enter Safe Sequence (space separated):"
read -a seq
echo "Safe Sequence is:"
for ((i=0;i<n;i++))
do
echo -n "${seq[$i]} "
done
echo
echo "System is in Safe State"
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNPRAVEEN P)~$ ./7bank.sh
Enter Number of Processes:
2
Enter Safe Sequence (space separated):
P1 P
Safe Sequence is:
P1 P
System is in Safe State
(kali@JOHNPRAVEEN P)~$
```